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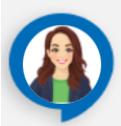
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Web based Order Management System

About the Project

Project Name	Web-based Order Management System (OMS)
Project Code	W-OMS01
Start Date	
End Date	
Managed	EdTech
Contract Type (FPP/T&M/Others)	
PM Name and email-id	Vidhyalakshmi
Domain	Manufacturing

Organization Vision:

ABC Power Equipment Inc. aims to become a leading manufacturer in the Power Equipment industry by leveraging technology to optimize operations, enhance customer satisfaction, and drive business growth.

Objectives :

SMART Objective for the Web-based Order Management System (OMS) Project:

Specific:

Develop and implement a comprehensive web-based Order Management System (OMS) for ABC Manufacturing Inc. to streamline and automate the order processing, bill of materials generation, purchase order creation, invoicing, and payment tracking processes.

Measurable:

- Reduce order processing time from an average of 5 days to 2 days.
- Achieve 98% invoice accuracy by reducing manual errors.
- Improve on-time delivery rate from 85% to 95%.
- Increase user adoption rate to 90% within 3 months of system deployment.

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inable:

- Allocate a dedicated project team consisting of a project manager, business analyst, UX/UI designer, 4 full-stack developers, 2 quality assurance engineers, and a DevOps engineer.
- Utilize modern web technologies and frameworks such as React, Node.js, and SQL Server to ensure scalability and performance.
- Follow Agile development methodologies to deliver incremental value and accommodate changing requirements.

Relevant:

- The OMS project aligns with ABC Manufacturing Inc.'s vision of leveraging technology to optimize operations, enhance customer satisfaction, and drive business growth.
- The system will address critical pain points, including manual order processing, lack of visibility, and inefficient payment reconciliation, which directly impact the company's operational efficiency and financial performance.

Time-bound:

- Complete the development and implementation of the Order Management System within 18 weeks (4.5 months) from the project kick-off date.
- Deliver the system incrementally, with the following epic-level timelines:
 - Order Management Epic: 4 weeks
 - BOM Management Epic: 3 weeks
 - Purchase Order Management Epic: 4 weeks
 - Invoice and Payment Management Epic: 3 weeks
 - Reporting and Analytics Epic: 2 weeks
 - User Management and Security Epic: 2 weeks

By achieving these specific, measurable, attainable, relevant, and time-bound objectives, ABC Manufacturing Inc. will significantly enhance its order management process, improve operational efficiency, and drive business growth through the successful implementation of the Web-based Management System.

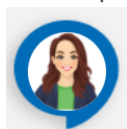
Metric for Report (W-OMS)	Operational Definition	Target
On-time Delivery	Percentage of orders delivered within the promised date	95%
Order Processing Time	Average time taken from order placement to shipment	2 days
Invoice Accuracy	Percentage of invoices generated without errors	98%
Payment Collection Time	Average time taken to collect payment after invoicing	30 days
User Adoption Rate	Percentage of users actively using the system	90%
System Uptime	Percentage of time the system is available and operational	99.9%

Existing Challenges : <Challenges which are faced by client/ project>

1. Manual and time-consuming order processing leading to delays and errors.
2. In-accurate Bill of Materials; with in-accurate part-no is getting generated for the order.
3. Lack of visibility and tracking of purchase orders and invoices.
4. Inefficient payment reconciliation and reporting.
5. Need for a centralized system with role-based access control.

Need for adopting agile : <Qualitative/ quantitative>

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Key Agile Tenet	Quantitative Benefits	Qualitative Benefits
Customer collaboration over contract negotiation	– Increase in customer satisfaction ratings by 25% due to regular feedback incorporation and alignment with customer needs. – Reduction in change requests and rework by 30% by actively involving customers throughout the development process.	– Enhanced trust and long-term partnerships with customers through transparent communication and collaboration. – Improved understanding of customer requirements, leading to the development of more relevant and valuable products.
Responding to change over following a plan	– Improved time-to-market by 20% through the ability to adapt to changing market conditions and customer requirements quickly. – Reduced waste and unused features by 15% by focusing on delivering the most valuable functionalities first and adjusting plans based on feedback.	– Increased agility and flexibility to respond to new opportunities, emerging technologies, and competitive threats. – Foster a culture of continuous improvement and innovation by embracing change and learning from feedback.
Working software over comprehensive documentation	– Accelerated delivery of working software increments by 25% by prioritizing functional code over extensive documentation. – Reduced documentation maintenance costs by 20% by keeping documentation lean and focused on essential information.	– Increased customer confidence and satisfaction by delivering tangible, working software early and frequently. – Improved collaboration and knowledge sharing among team members through face-to-face communication and working software demos.
Individuals and interactions over processes and tools	– Increased team productivity by 15% by fostering open communication, collaboration, and self-organization. – Reduced employee turnover rate by 10% by promoting a supportive and empowering work environment.	– Enhanced problem-solving capabilities and creativity through the exchange of ideas and diverse perspectives. – Improved team morale, motivation, and ownership by valuing individuals and their contributions over rigid processes and tools.

Type of work : <development >

Scope of the project:

The web-based Order Management System will include the following modules:

1. Order Booking Module
2. Bill of Material (BOM) Module
3. Purchase Order Module
4. Invoice Generation Module
5. Payment Module
6. Reporting and Analytics Module
7. User Management and Access Control Module

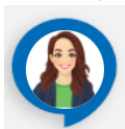
Available options:

1. Custom development of the entire system from scratch.
2. Utilizing an off-the-shelf order management solution and customizing it to meet specific requirements.
3. Integrating existing systems with a new custom-built order management module.

Epics in Scope:

1. Order Management
 - User story: As a sales representative, I want to create and manage sales orders so that I can efficiently process customer orders.
2. BOM Management
 - User story: As a production manager, I want to generate detailed bill of materials based on product configurations so that I can accurately plan and manage production.
3. Purchase Order Management

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- User story: As a purchasing officer, I want to create and track purchase orders for required components so that I can ensure timely procurement and maintain vendor relationships.

4. Invoice and Payment Management

- User story: As a finance manager, I want to generate invoices and track payments so that I can maintain accurate financial records and ensure timely payment collection.

5. Reporting and Analytics

- User story: As a business analyst, I want to access comprehensive reports and dashboards so that I can gain insights into order status, inventory levels, and financial performance.

6. User Management and Security

- User story: As an administrator, I want to manage user roles and access controls so that I can ensure data security and compliance.

Scope and Duration for Development:

- Order Management Epic:
 - Scope: Develop the order booking module, including order creation, editing, and tracking capabilities.
 - Duration: 4 weeks
- BOM Management Epic:
 - Scope: Develop the BOM module to generate detailed bill of materials based on product configurations.
 - Duration: 3 weeks
- Purchase Order Management Epic:
 - Scope: Develop the purchase order module to create, track, and manage purchase orders.
 - Duration: 4 weeks
- Invoice and Payment Management Epic:
 - Scope: Develop the invoice generation and payment tracking modules.
 - Duration: 3 weeks
- Reporting and Analytics Epic:
 - Scope: Develop comprehensive reports and dashboards for order status, inventory, and financial performance.
 - Duration: 2 weeks
- User Management and Security Epic:
 - Scope: Implement user role management and access control features.
 - Duration: 2 weeks

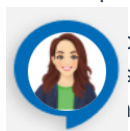
About Client

ABC Power Equipment Inc. is a leading manufacturer of small and large power supply devices for both industrial and residential use. Founded in 1985, the company has established itself as a trusted name in the power equipment industry, known for its innovative designs, high-quality products, and exceptional customer service.

ABC Power Equipment Inc. has a diverse product portfolio, which includes:

1. Industrial-grade power supplies:
 - Uninterruptible Power Supply (UPS) systems
 - Switchgear and switchboards
 - Transformers and substations
 - Motor control centers
 - Industrial batteries and chargers
2. Residential power solutions:
 - Portable generators
 - Standby generators
 - Inverters and converters
 - Solar power systems
 - Battery backup systems

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Company has a strong presence in both domestic and international markets, with manufacturing facilities located in the United States, China, and Germany. ABC Power Equipment Inc. serves a wide range of customers, including utility companies, data centers, healthcare facilities, manufacturing plants, commercial buildings, and residential households.

ABC Power Equipment Inc. has a reputation for delivering reliable, efficient, and sustainable power solutions. The company invests heavily in research and development to stay at the forefront of technological advancements in the power equipment industry. They have a dedicated team of engineers, designers, and technical experts who work collaboratively to develop cutting-edge products that meet the evolving needs of their customers.

The company places a strong emphasis on quality control and adheres to stringent industry standards and regulations. Their products undergo rigorous testing and certification processes to ensure optimal performance, safety, and reliability.

ABC Power Equipment Inc. is committed to providing exceptional customer support and has a global network of distributors, dealers, and service centers to ensure prompt and efficient after-sales service. The company also offers comprehensive training programs and technical support to help customers maximize the performance and longevity of their power equipment.

With a strong financial position, a well-established brand, and a customer-centric approach, ABC Power Equipment Inc. is well-positioned to capitalize on the growing demand for reliable and sustainable power solutions in both industrial and residential markets.

Project Technology

- Front-end: React, HTML, CSS, JavaScript
- Back-end: Node.js, Express.js
- Database: Oracle SQL
- DevOps: AWS, Jenkins, Docker, Kubernetes

Team Details

Here’s a breakdown of the team roles, responsibilities, quantity, location, organizational structure, and reporting hierarchy for the web-based Order Management System (OMS) project:

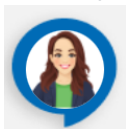
Sr. No.	Role	Quantity	Responsibilities	Key Tasks	Reports To	Tools Used
1	Project Steering Committee	-	- Provide overall project guidance and strategic direction - Approve project charter, budget, and timeline - Make high-level decisions and resolve escalated issues	- Conduct regular project review meetings - Approve project milestones and deliverables - Resolve project-related issues and risks	-	- Project management software (e.g., Microsoft Project, Smartsheet) - Video conferencing tools (e.g., Zoom, Microsoft Teams)
1.1	Project Manager	1	- Overall project planning, execution, and monitoring - Stakeholder management and communication - Risk and issue management - Ensuring project deliverables are met within the defined scope, timeline, and budget	- Develop project plan and schedule - Conduct stakeholder meetings and manage communication - Monitor project progress and adjust plans as needed - Identify and mitigate project risks and issues	Project Steering Committee	- Project management software (e.g., JIRA, Asana) - Collaboration tools (e.g., Slack, Microsoft Teams) - Document management tools (e.g., Google Drive, Confluence)

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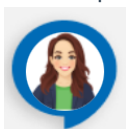
Sr. No.	Role	Quantity	Responsibilities	Key Tasks	Reports To	Tools Used
1.1.1	Business Analyst	1	– Gathering and documenting business requirements – Conducting stakeholder interviews and workshops – Defining user stories and acceptance criteria – Collaborating with the development team to ensure requirements are understood and implemented correctly	– Elicit and document business requirements – Create user stories and acceptance criteria – Facilitate requirements gathering workshops – Collaborate with the development team to clarify requirements	Project Manager	– Requirements management tools (e.g., JIRA, Confluence) – Wireframing and prototyping tools (e.g., Balsamiq, Sketch) – Collaboration tools (e.g., Slack, Microsoft Teams)
1.1.2	UX/UI Designer	1	– Designing intuitive and visually appealing user interfaces – Creating wireframes, mockups, and prototypes – Conducting user research and usability testing – Collaborating with the development team to ensure design consistency and optimal user experience	– Create wireframes, mockups, and prototypes – Conduct user research and usability testing – Design user interfaces and user flows – Collaborate with the development team to implement designs	Project Manager	– Design tools (e.g., Sketch, Adobe XD, Figma) – Prototyping tools (e.g., InVision, Marvel) – User research and testing tools (e.g., UserTesting, UsabilityHub)

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Sr. No.	Role	Quantity	Responsibilities	Key Tasks	Reports To	Tools Used
1.1.3	Technical Lead	1	– Oversee both the Onsite and Offshore development teams – Provide technical guidance and mentorship to the development team – Ensure adherence to coding standards and best practices – Collaborate with the Project Manager and Business Analyst to align technical implementation with business requirements	– Define technical architecture and design – Provide technical leadership and guidance to the development team – Conduct code reviews and ensure code quality – Collaborate with the Project Manager and Business Analyst to ensure technical feasibility and alignment with requirements	Project Manager	– Development IDEs (e.g., Visual Studio, IntelliJ) – Version control tools (e.g., Git, GitHub) – Continuous integration and deployment tools (e.g., Jenkins, GitLab CI/CD) – Code review tools (e.g., Gerrit, Crucible)
1.1.3.1	Full-stack Developers	4	– Developing front-end and back-end components of the OMS – Implementing user stories and features based on the defined requirements – Collaborating with the QA team to ensure code quality and bug-free deliverables – Participating in code reviews and knowledge sharing sessions	– Develop front-end and back-end components – Implement user stories and features – Write unit tests and integrate with CI/CD pipeline – Participate in code reviews and collaborate with the QA team	Technical Lead	– Development IDEs (e.g., Visual Studio, IntelliJ) – Programming languages (e.g., Java, JavaScript, Python) – Front-end frameworks (e.g., React, Angular) – Back-end frameworks (e.g., Node.js, Spring Boot) – Databases (e.g., MySQL, MongoDB) – Testing frameworks (e.g., JUnit, Mocha)

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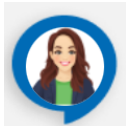


Sr. No.	Role	Quantity	Responsibilities	Key Tasks	Reports To	Tools Used
1.1.4	QA Lead	1	- Oversee the Quality Assurance Engineers - Define and implement the testing strategy - Ensure comprehensive test coverage and timely defect resolution - Collaborate with the Project Manager and Technical Lead to maintain high-quality deliverables	- Define testing strategy and test plans - Oversee test case development and execution - Ensure defect tracking and resolution - Collaborate with the development team to ensure quality and timely delivery	Project Manager	- Test management tools (e.g., JIRA, TestRail) - Automated testing tools (e.g., Selenium, Appium) - Defect tracking tools (e.g., JIRA, Bugzilla) - Collaboration tools (e.g., Slack, Microsoft Teams)
1.1.4.1	Quality Assurance Engineers	2	- Developing and executing test cases and test scenarios - Performing manual and automated testing - Reporting and tracking bugs and issues - Collaborating with the development team to ensure timely resolution of defects	- Develop and execute test cases - Perform manual and automated testing - Report and track defects - Collaborate with the development team to resolve defects	QA Lead	- Test management tools (e.g., JIRA, TestRail) - Automated testing tools (e.g., Selenium, Appium) - Defect tracking tools (e.g., JIRA, Bugzilla) - Collaboration tools (e.g., Slack, Microsoft Teams)
1.1.5	DevOps Engineer	1	- Setting up and maintaining the development, testing, and production environments - Implementing CI/CD pipelines for automated build, testing, and deployment - Monitoring system performance and troubleshooting issues - Ensuring high availability and scalability of the OMS	- Set up and maintain development, testing, and production environments - Implement CI/CD pipelines - Monitor system performance and troubleshoot issues - Ensure high availability and scalability of the infrastructure	Technical Lead	- Cloud platforms (e.g., AWS, Azure) - Configuration management tools (e.g., Ansible, Puppet) - Containerization tools (e.g., Docker, Kubernetes) - Monitoring and logging tools (e.g., Prometheus, ELK stack) - Collaboration tools (e.g., Slack, Microsoft Teams)

IT RESOURCES:

Development Environment: Cloud-based infrastructure (AWS or Azure)

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- Source Control: Git
- Issue Tracking and Project Management: JIRA
- Continuous Integration/Continuous Deployment (CI/CD): Jenkins

- Code Quality and Analysis: SonarQube
- Collaboration Tools: Teams

Other Resources:

- Training materials for end-users
- User acceptance testing environment

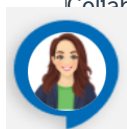
Metrics

Metric Level	Metric Name	Operational Definition	Drives
Strategic	Return on Investment (ROI)	Measure the financial return generated by the OMS project in relation to the total investment made.	On-Time Delivery, Order Processing Time, Invoice Accuracy, System Adoption Rate
	Customer Satisfaction Score	Assess the level of satisfaction among end-users and stakeholders with the implemented OMS solution.	On-Time Delivery, Order Processing Time, Invoice Accuracy, System Adoption Rate
Tactical	On-Time Delivery	Percentage of orders delivered within the promised date.	Velocity, Sprint Burndown
	Order Processing Time	Average time taken from order placement to shipment.	Cycle Time
	Invoice Accuracy	Percentage of invoices generated without errors.	Defect Density, Code Coverage
	System Adoption Rate	Percentage of intended users actively using the OMS.	User Story Acceptance Rate, Sprint Goal Success Rate
Operational	Velocity	Average number of story points completed per sprint.	On-Time Delivery
	Sprint Burndown	Measure of work completed against the sprint plan.	On-Time Delivery
	Cycle Time	Average time taken from task initiation to completion.	Order Processing Time
	Defect Density	Number of defects found per unit of work (e.g., story points or features).	Invoice Accuracy
	Code Coverage	Percentage of code covered by automated tests.	Invoice Accuracy
	User Story Acceptance Rate	Percentage of user stories accepted by the product owner in each sprint.	System Adoption Rate
	Sprint Goal Success Rate	Percentage of sprints where the sprint goal is fully achieved.	System Adoption Rate

TOOLS

Category	Tool Type	Tools
Project Management	Project Management	JIRA
Engineering Tools	Build & CI	Jenkins
	Code Review and Analysis	SonarQube
	Configuration	Ansible
	Construction	Visual Studio
	Defect Mgmt	JIRA
	Unit Testing	Jest
Collaboration Tools	Collaboration	Slack
Technology Stack	Front-end	React

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Category	Tool Type	Tools
	Back-end	Node.js, Express.js
	Database	SQL Server

Agile Practices

- Daily stand-up meetings
- Sprint planning and review meetings
- Retrospectives
- User story mapping
- Continuous integration and continuous deployment (CI/CD)
- Pair programming
- Test-driven development (TDD)

Sample Reporting from Tools:

- JIRA Dashboard:
- Sprint burndown chart
- Velocity chart
- Cumulative flow diagram
- Epic progress report
- Kanban Board:
- Visualization of work in progress
- Cycle time and lead time metrics
- Bottleneck identification

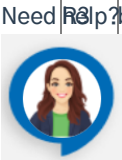
Best practices

1. Involve key stakeholders throughout the development process to ensure alignment with business requirements.
2. Prioritize user stories based on business value and deliver incrementally to gather early feedback.
3. Implement a robust CI/CD pipeline to enable frequent deployments and maintain high code quality.
4. Conduct regular code reviews and implement static code analysis to identify and address potential issues early.
5. Perform comprehensive testing, including unit testing, integration testing, and user acceptance testing.
6. Provide adequate training and support to end-users to ensure successful adoption of the system.
7. Continuously monitor system performance and gather user feedback to identify areas for improvement.

Risks, Issues, Lessons Learnt from Past

Table 1: Risk Register with Risk Response Strategies

Risk ID	Risk Description	Probability	Impact	Risk Rating	Risk Response Strategy	Risk Owner
R1	Scope creep due to unclear requirements	High	High	High	Clearly define and document requirements – Establish change control process –Regularly communicate with stakeholders	Project Manager
R2	Delays in development due to technical complexity	Medium	High	High	Break down complex tasks into smaller, manageable units – Allocate experienced resources to complex tasks – Provide necessary training and support	Technical Lead
R3	Integration issues between different modules	Medium	Medium	Medium	Define clear integration points and APIs – Conduct regular integration testing – Establish communication channels between development teams	Technical Lead



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Risk ID	Risk Description	Probability	Impact	Risk Rating	Risk Response Strategy	Risk Owner
R4	Resource unavailability due to competing projects	Medium	Medium	Medium	Identify and allocate dedicated resources – Establish resource leveling and capacity planning – Communicate resource requirements to management	Project Manager
R5	Delays in customer feedback and approval	Low	Medium	Low	Establish regular communication with customers – Set clear expectations for feedback timelines – Implement a feedback tracking system	Business Analyst

Table 2: Historical Issues and Challenges to Consider

Issue ID	Issue Description	Impact	Resolution
I1	Lack of user involvement during requirements gathering	High	Engage users throughout the development process – Conduct regular user feedback sessions – Implement user acceptance testing
I2	Inadequate testing leading to production defects	High	Develop comprehensive test plans – Allocate sufficient time and resources for testing – Implement automated testing wherever possible
I3	Poor code quality and maintainability	Medium	Establish coding standards and best practices – Conduct regular code reviews – Implement static code analysis tools
I4	Communication breakdowns between teams	Medium	Establish clear communication channels – Conduct regular stand-up meetings and status updates – Encourage cross-functional collaboration
I5	Insufficient training and documentation	Low	Develop comprehensive user manuals and guides – Provide training sessions for users and support staff – Maintain up-to-date documentation

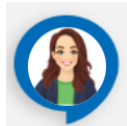
Table 3: Lessons Learned from Past Projects

Lesson ID	Lesson Description	Recommendations
L1	Importance of clear and detailed requirements	Allocate sufficient time for requirements gathering – Involve all relevant stakeholders in requirements discussions – Use visual aids like wireframes and prototypes
L2	Benefits of continuous integration and deployment	Implement a robust CI/CD pipeline – Automate build, testing, and deployment processes – Establish a culture of frequent releases and feedback
L3	Value of effective communication and collaboration	Encourage open and transparent communication – Use collaboration tools like Slack and JIRA – Foster a culture of teamwork and shared responsibility
L4	Significance of performance testing and optimization	Conduct regular performance testing – Identify and optimize performance bottlenecks – Monitor production performance metrics
L5	Importance of change management and user adoption	Develop a comprehensive management plan – Engage users early and provide adequate training – Celebrate successes and milestones to maintain momentum

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